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## Online Decision-Making Tool for Active Shade Management

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Shade is a critical urban design feature for mitigating heat stress, particularly in hot, arid cities. Its impact on pedestrians is highly localized, site-specific, and dependent on multiple factors, including shade type, orientation, and ground cover. As cities strive to address "shade deserts"—urban areas lacking adequate shade infrastructure—decision-support tools are needed to inform optimal shade placement and design. We developed a web-based decision-support tool that generates shade performance curves to evaluate the cooling efficacy of various shade types. Users can customize parameters such as shade type, size, location, and date to assess their effectiveness in reducing heat load. The tool displays a rendered hemispherical view of the shade structure and a graph illustrating the hourly reduction in mean radiant temperature ( $\Delta\text{MRT}$ ) compared to a sun-exposed location. We validated the tool using MaRTy, a human-biometeorological cart, for hot, sunny summer days in Phoenix, Arizona. Sun-exposed locations had an RMSE of 5.9°C and an MBE of -4.2°C. Umbrella shade was underestimated systematically (MBE = -4.39°C) with an RMSE of 6.0°C. Tree shade was modeled with an RMSE of 5.1°C and a lower bias (-2.0°C). Shade performance varied by ground type, with concrete yielding the best accuracy (RMSE = 4.0°C,  $d = 0.86$ ), while grass exhibited greater variability. A sensitivity analysis was conducted to investigate shade performance across shade types, seasons, and times of day. The web-based decision-making tool will allow users to assess the heat load impacts of customized, site-specific shade interventions for active shade management, offering planners and designers a robust tool for strategic shade implementation.